
EE133 Solid State Electronics

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Semiconductor Device Physics

Why This Course Matters More Than You Think

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The Big Misconception

- Most people think computing progress comes from software.
- Reality: software improvements are unnoticeable compared to device physics – unless you use a log scale.
- $>10^9\times$ performance gain in computing capability since 1970.
- $\sim 10^8\times$ due to transistor scaling and device physics breakthroughs.

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Course Overview and Motivation

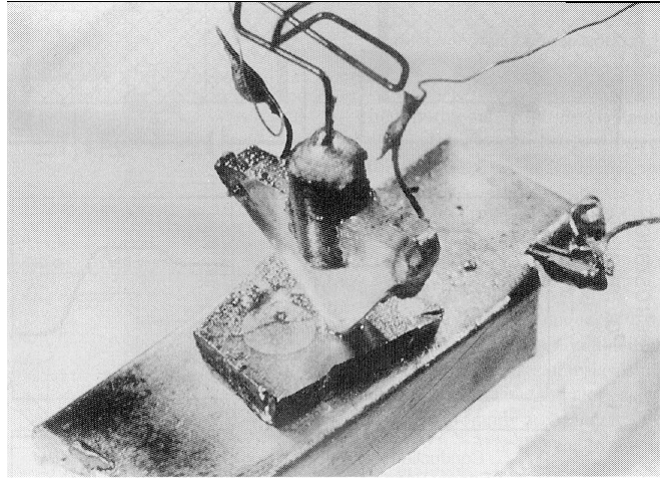
- Purpose of EE133: Understand the operation of solid state (Si) devices like diodes and transistors
- Physics of semiconductor devices is multi-disciplinary
 - Electrical engineering (Circuits and systems)
 - Physics (Device operation – solid state physics)
 - Chemistry / Chem. Eng. (Device fabrication)
 - Biology (Bio applications)
- Most interdisciplinary area within EE.
- Overview of solid state devices
 - History
 - Industry
 - Research at UCR

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History of Computers

The Transistor Revolution



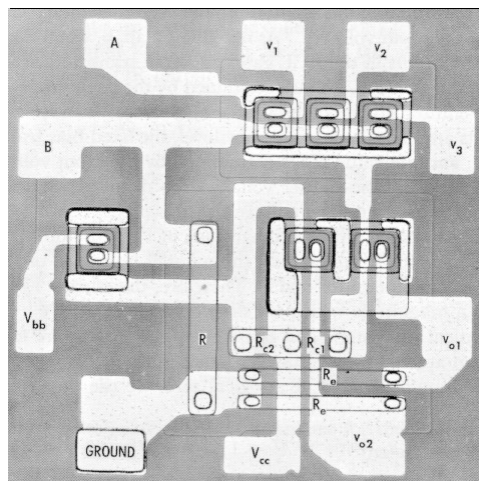
First transistor
Bell Labs, 1948

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History of Computers

The First Integrated Circuits



*Bipolar logic
1960's*

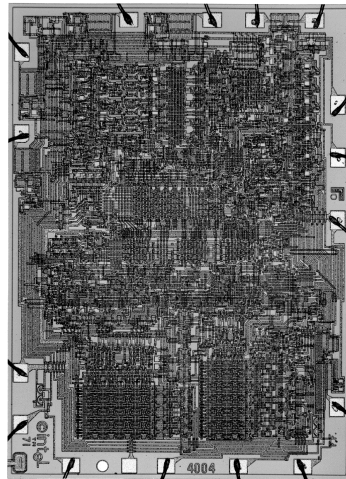
ECL 3-input Gate
Motorola 1966

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History of Computers

Intel 4004 Micro-Processor

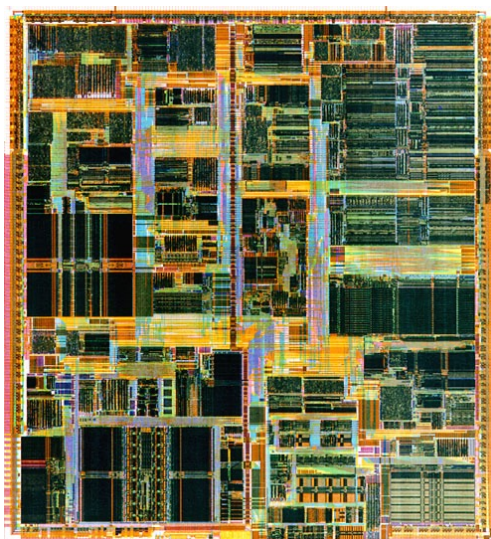


1971
1000 transistors
1 MHz operation

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Intel Pentium (IV) microprocessor



2000
500M transistors
3GHz operation

3000 times faster clock speeds
than in 1971.

500,000 times more transistors
than in 1971.

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Moore's Law

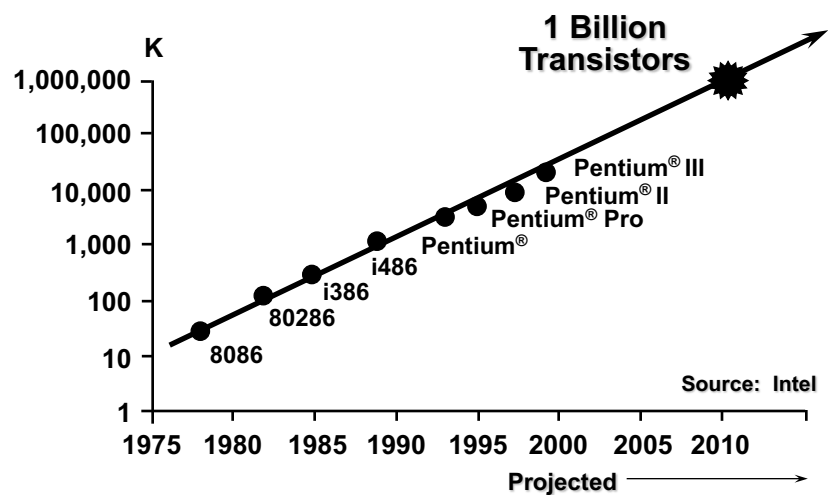
- In 1965, Gordon Moore noted that the number of transistors on a chip doubled every 18 to 24 months.
- He made a prediction that semiconductor technology will double its effectiveness every 18 months.

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Moore's Law

Transistor Counts

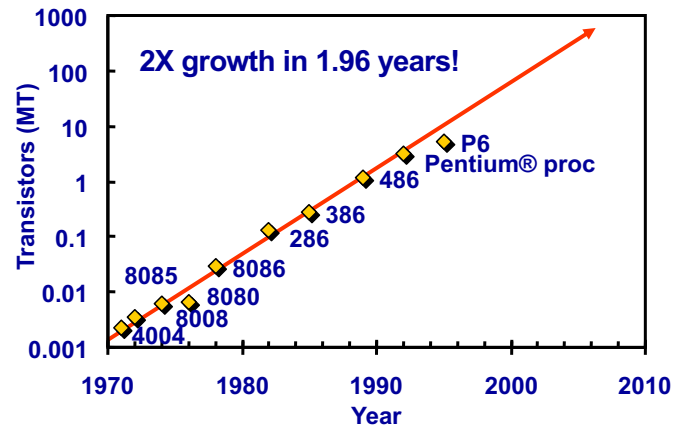


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Moore's Law

Moore's law in Microprocessors



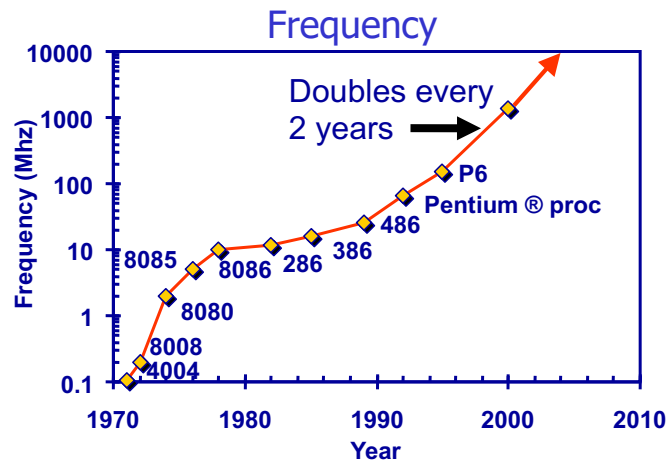
Transistors on Lead Microprocessors double every 2 years

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Courtesy, Intel

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Moore's Law



Lead Microprocessors frequency doubles every 2 years

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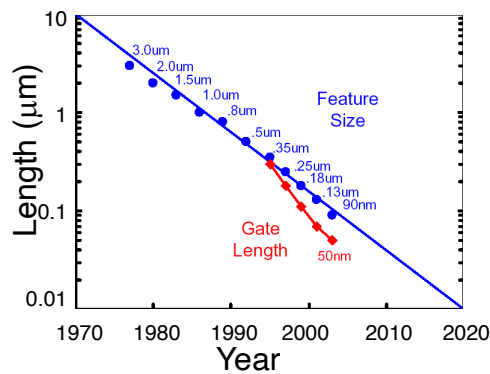
Courtesy, Intel

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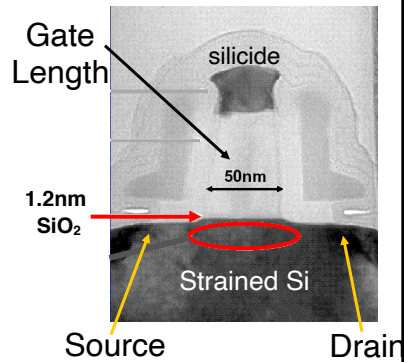
Moore's Law

Intel: August 2002
Commercial Electronics Already Nanoelectronics

Unexpected Acceleration of Moore's Law



State of Art MOSFET



www.intel.com/research/silicon/90nm_press_briefing-technical.htm

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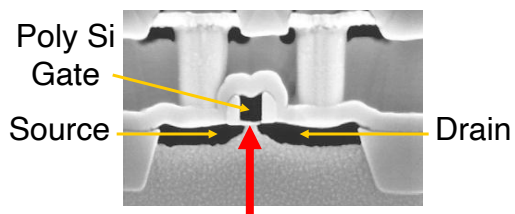
Material Limits

- silicon
- metal interconnects
- interlevel dielectrics
- **gate dielectric**

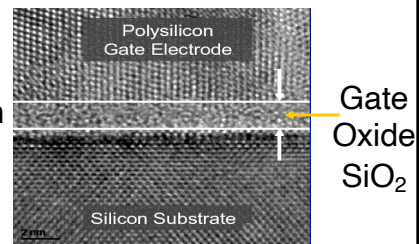
Min thickness?

Quantum mechanical
tunneling leakage current

Si MOSFET



1.2 nm

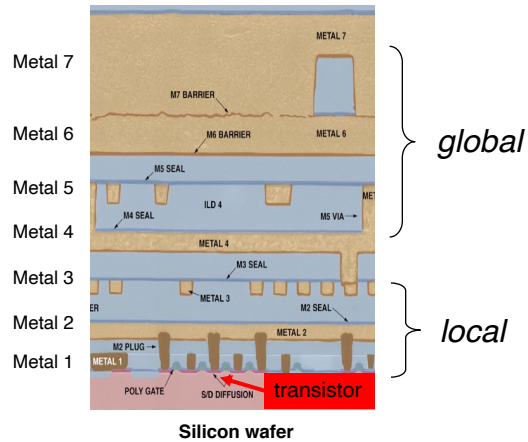


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Circuit Limits

- Speed
- Power

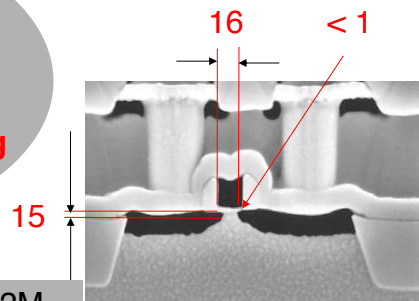


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Practical and Economic Limits

- lithography
- etching
- doping, etc.
- **atomic scale manufacturing**



1967 Cost of a Silicon Fab: \$ 2M
 2002 Cost of a Silicon Fab: ~ \$ 3B
2015 Cost of a Silicon Fab: ~\$100B

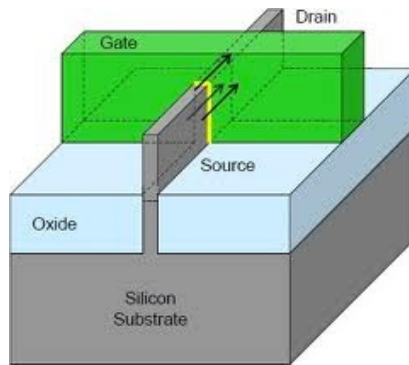
MOSFET

All dimensions in units of the Silicon lattice constant, 5.4 Å

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FinFETs and Hi-K Dielectrics

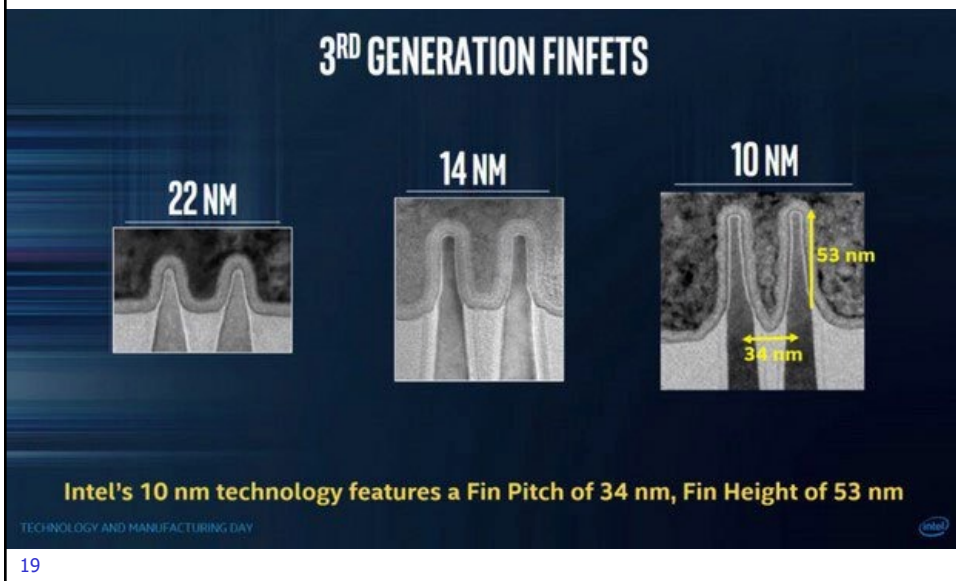


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FinFETs

Intel's Cannonlake's 10 nm finFET node. (March 2017)

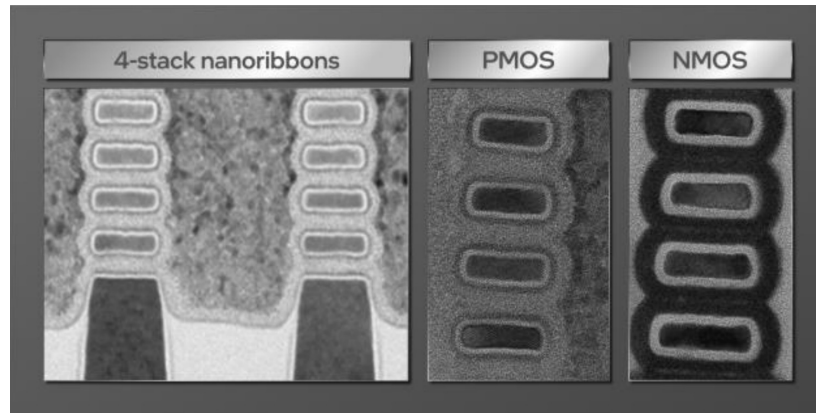


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RibbonFETs

Intel IEDM Dec. 2022

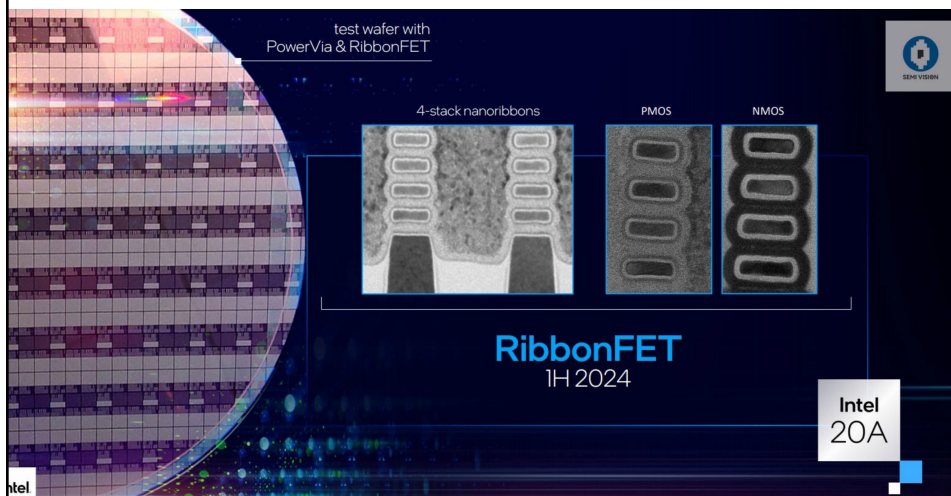


A. B. Kelleher, "Celebrating 75 years of the transistor A look at the evolution of Moore's Law innovation," IEDM (Dec. 2022).

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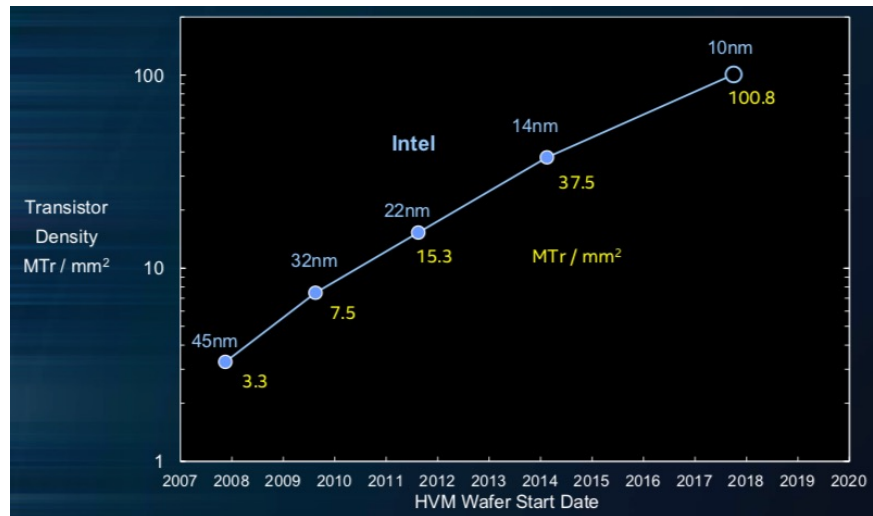
2023 IEDM



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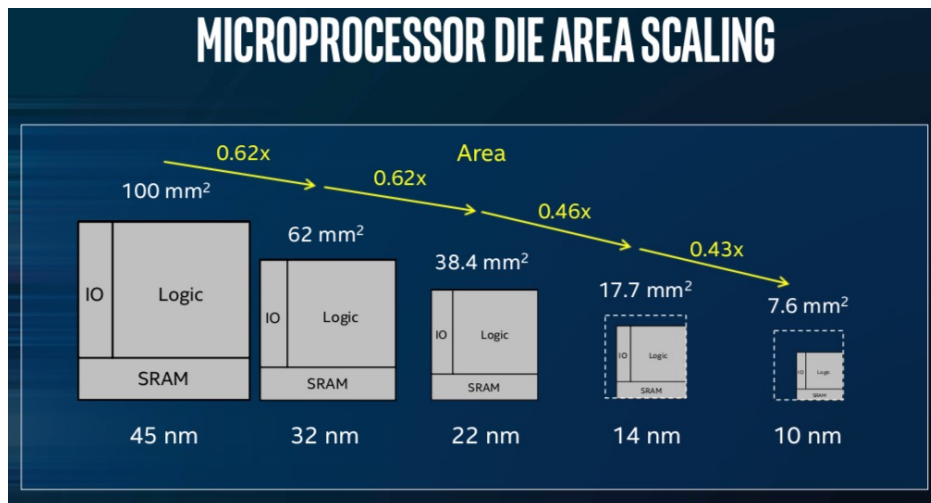
Intel 2017



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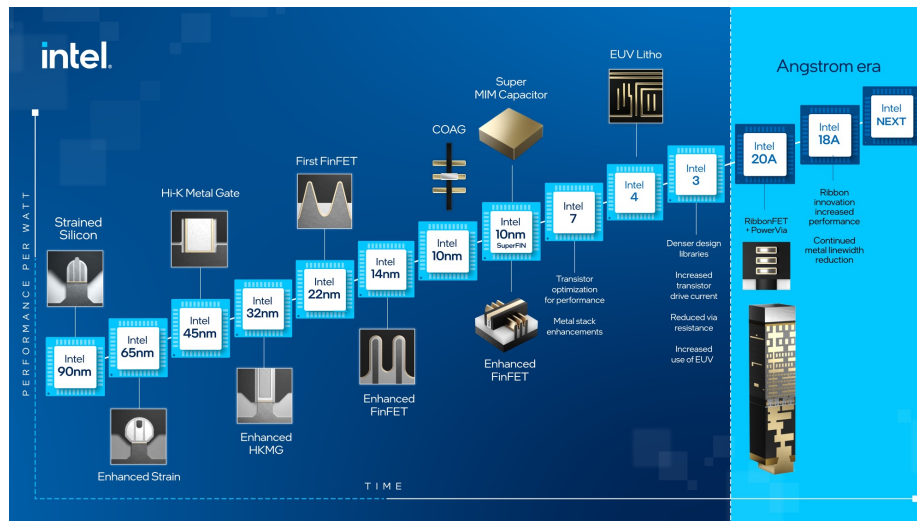
CPU die size (Intel 2017)



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Intel Roadmap



A. Kelleher, "Moore's Law – Now and in the Future," Intel newsroom, Feb. 2022.

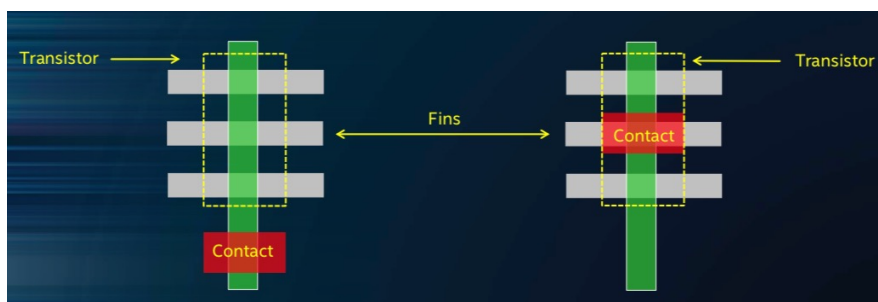
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FinFET Transistor cell (Intel 2017)

Very large scale integration (VLSI) Layout: EE/CS 168

Chip Designers – traditionally at the top of the technical pyramid in industry.



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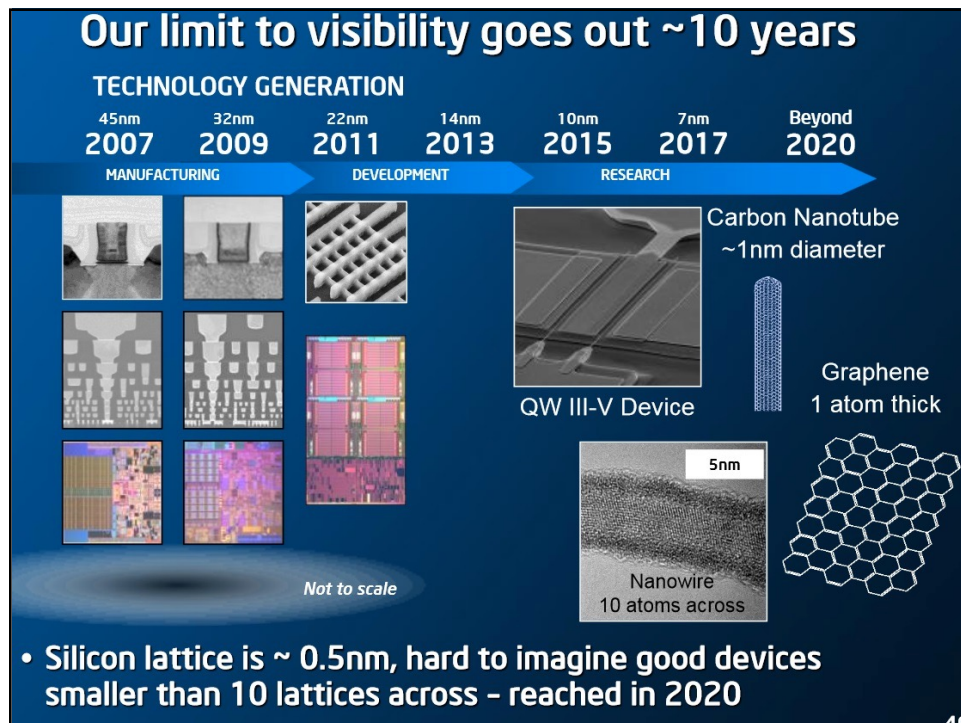
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The Big Picture – 12" Si Wafer



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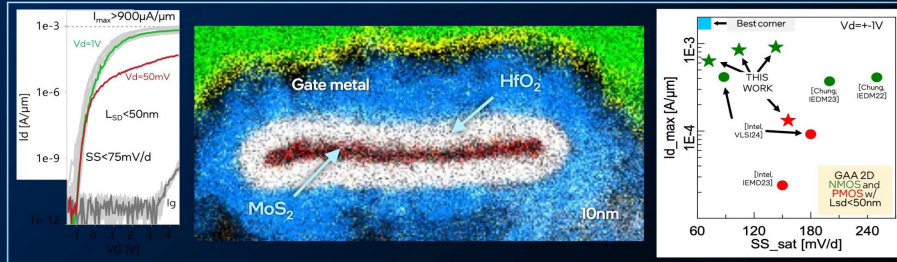
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Accelerating GAA beyond CFET - 2D NMOS & PMOS Transistors

Achieved best-in-class 2D GAA transistor with a specific focus on gate oxide module development.



- Intel advances two-dimensional (2D) transition metal dichalcogenide (TMD) semiconductors for future advanced transistor nodes.

- 2X NMOS current improvement over best published result.



- First GAA 2D transistor with good subthreshold slope (<75mV/d) using ALD HfO2 process.

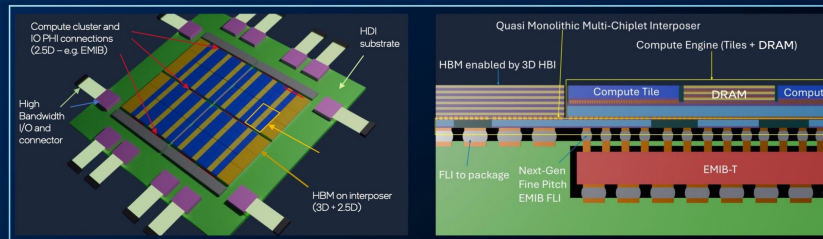
intel foundry

Technical Paper: Gate oxide module development for scaled GAA 2D FETs ... (Authors: Wouter Montelmann, Pratyush Buragohain, Ande Kitamura)

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Tomorrow's Modular and Compute Systems: an IEDM Invited Talk



Three key packaging solutions needed to deliver AI systems at scale:

- Advanced memory integration to eliminate capacity, bandwidth and latency bottlenecks,
- Utilization of hybrid bonding interconnect pitch scaling to access power-efficient, bandwidth density between heterogeneous components, and;
- Modularity system expansion with connectivity solutions to avoid networked latency and bandwidth limitations.



intel foundry

Technical paper: Tomorrow's Modular & Scalable Compute Systems (Authors: Johanna Swan, Aleksander Aleksov, George Dogiamis, Adel Elsherbini)

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CHIPS and Science Act of 2022

- The bill appropriates \$54.2 billion for subsidies to build chip plants in the U.S. and support U.S. chip research and development.
- It provides a 25% tax credit for building and equipping U.S. chip plants which is estimated to provide another \$24.3 billion in support.
- It significantly increases authorizations for federal science and technology research and development programs, specifically authorizing about \$174 billion through FY 2027 to support the nation's science and technology base — including for interagency programs to boost technological innovation and help translate federally funded research to commercial applications.

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EE133

- EE 133 is the entry course into this interdisciplinary area.

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